CONTROLLING QUANTUM FLUX WITH MEASUREMENT

and a Quantum Maxwell Daemon...

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OUTLINE

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INTRODUCTION

RECALL WHAT MICHEL SAID

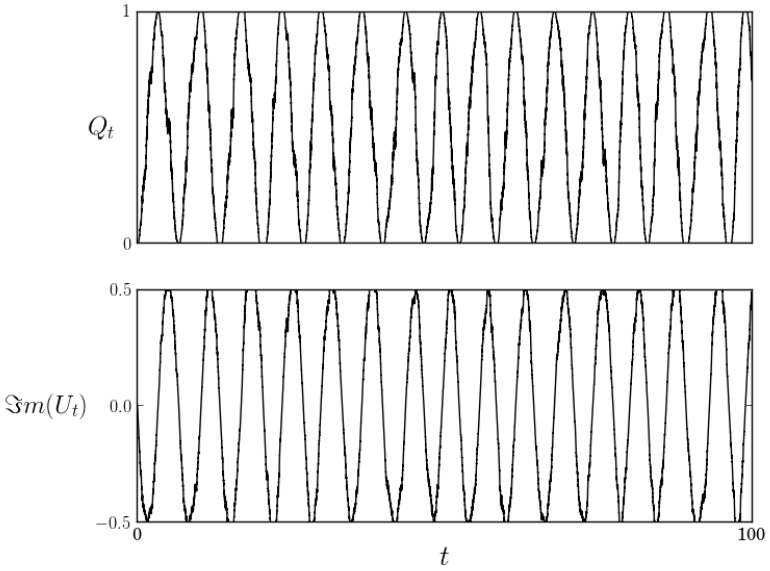
Quantum jumps

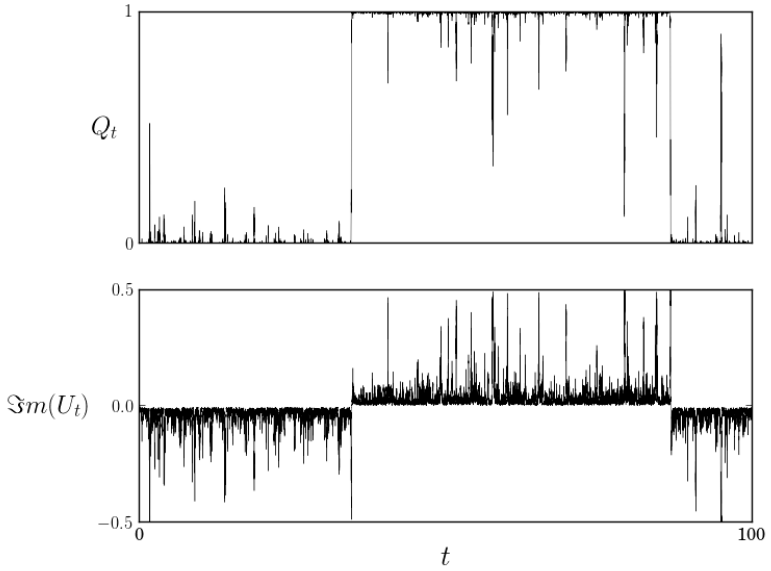
The density matrix of a quantum system that is continuously measured evolves like a finite state Markov process, i.e. it **jumps between pointer states**, in the strong measurement limit (and if the density matrix is conditioned on the measurement results).

- Universal signature : spikes (not needed now)
- Different from the Poissonian unravelling of a master equation
- The jump rates can be ”easily” computed

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"Hamiltonian" jumps





Small γ

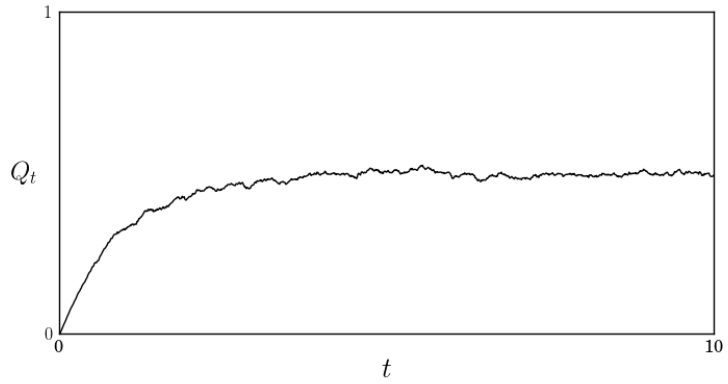
Large γ

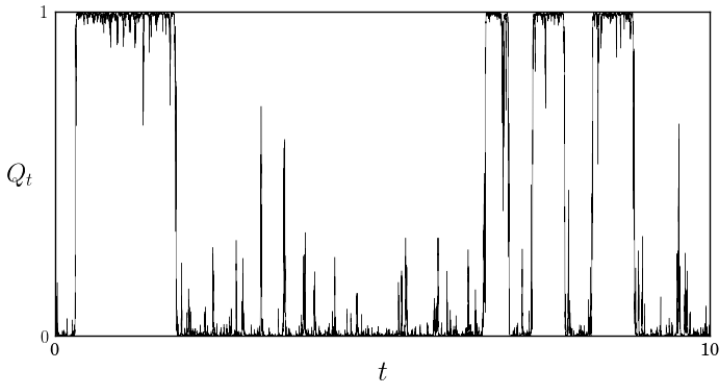
Jump rates $\propto \gamma^{-2}$ $\rightarrow$ Zeno frozen

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"*Thermal*" jumps





Small γ

Large γ

Jump rates **independent** of γ

$\rightarrow$ not Zeno frozen

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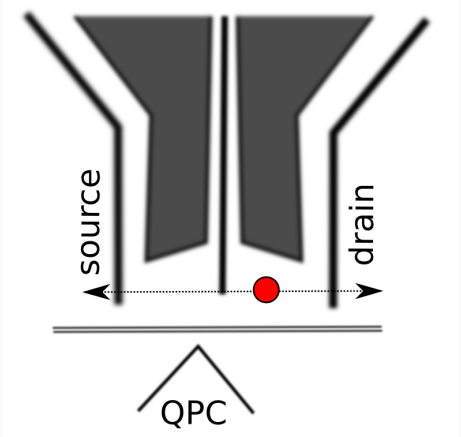
RECALL WHAT MICHEL SAID

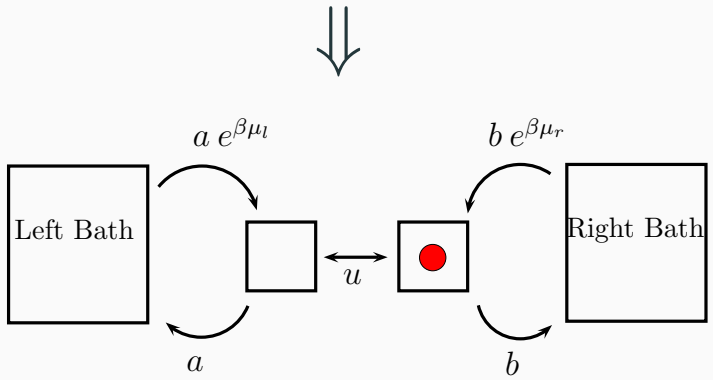
Study a model where the two jumps compete !

MODEL

INSPIRATION

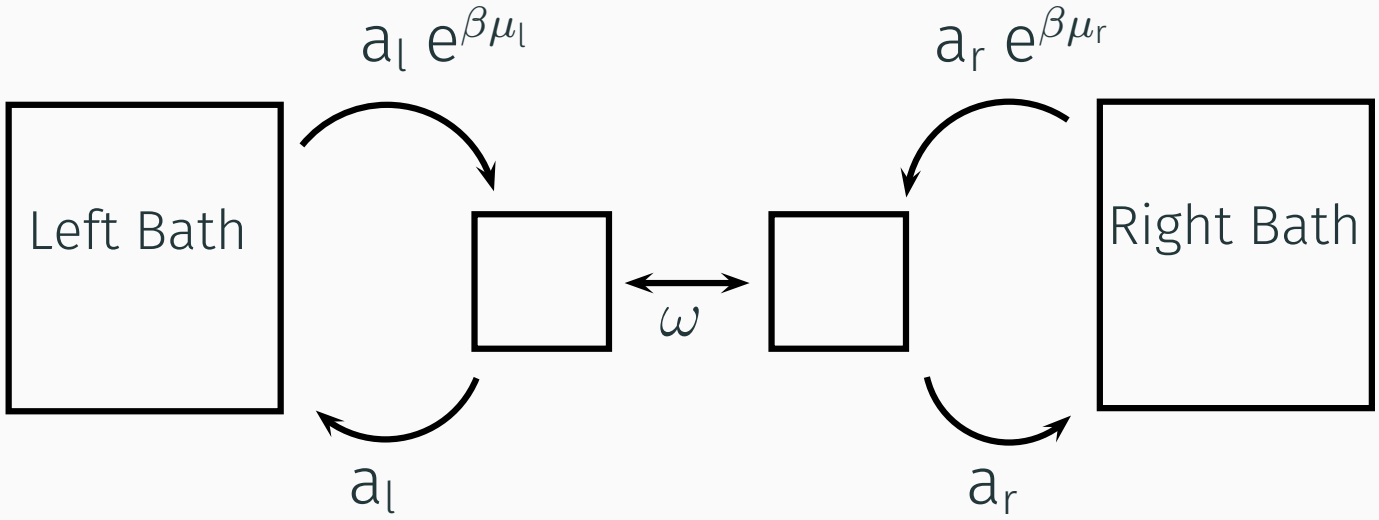
Physical system in mind from: **Küng et al.** Test of the fluctuation theorem for single-electron transport, J. Appl. Phys. (2013)





SCHEMATICS OF OUR MODEL

3-level system : $\mathcal{H} = \mathrm{Vect}(|0\rangle, |\mathrm{L}\rangle, |\mathrm{R}\rangle)$

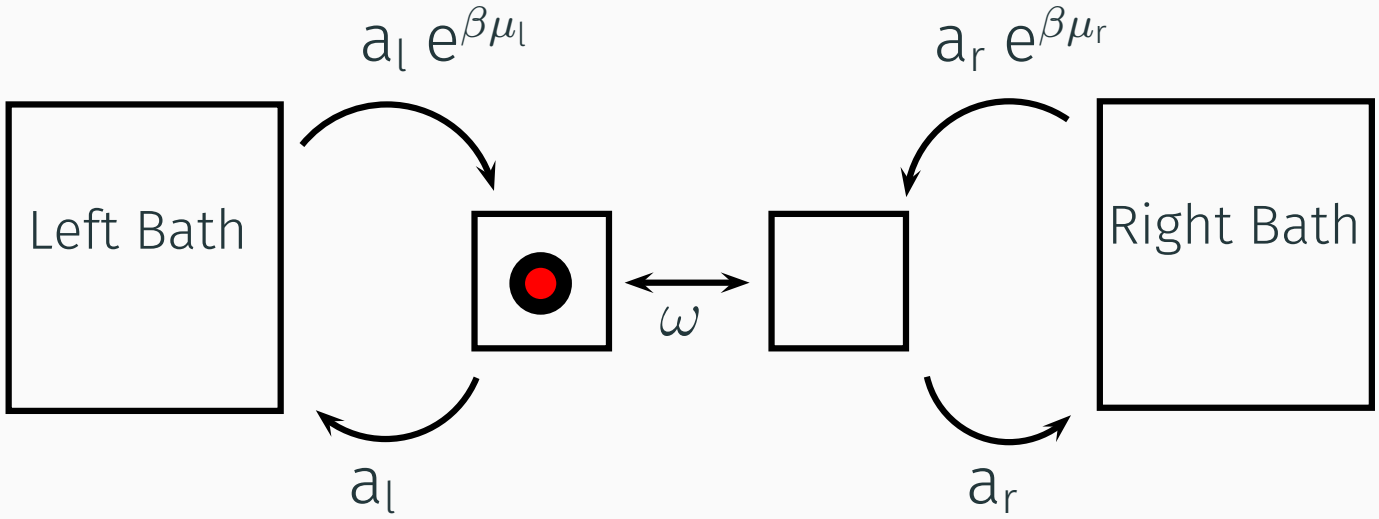


$$|0\rangle$$

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SCHEMATICS OF OUR MODEL

3-level system : $\mathcal{H} = \mathrm{Vect}(|0\rangle, |\mathrm{L}\rangle, |\mathrm{R}\rangle)$

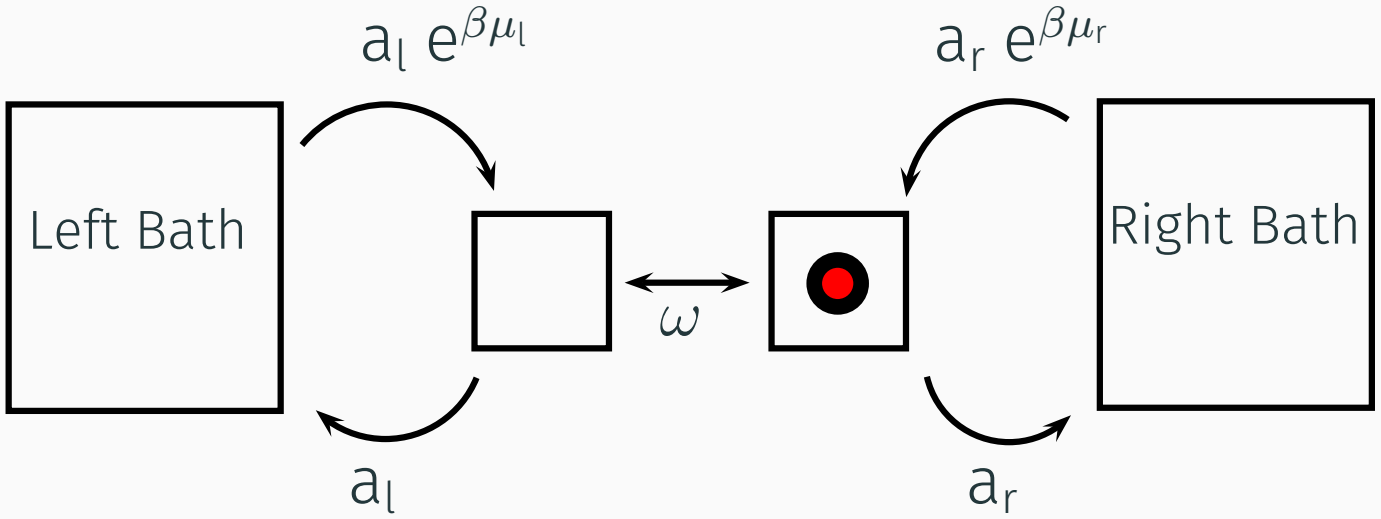


$|\mathrm{L}\rangle$

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SCHEMATICS OF OUR MODEL

3-level system : $\mathcal{H} = \mathrm{Vect}(|0\rangle, |\mathrm{L}\rangle, |\mathrm{R}\rangle)$

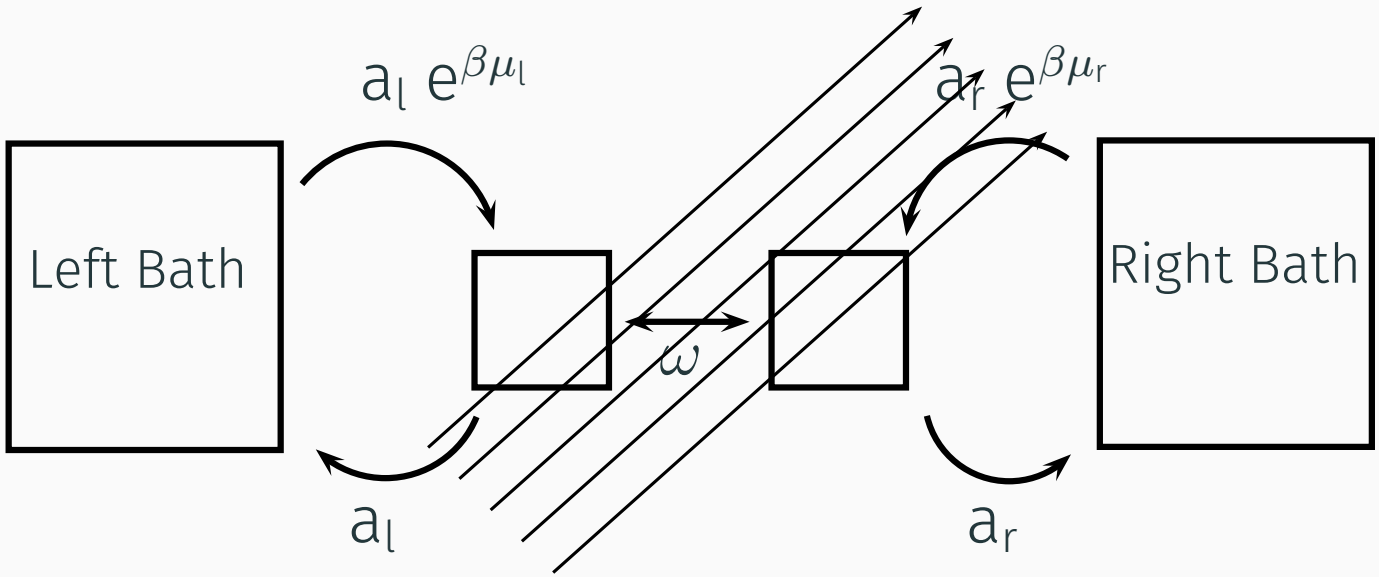


$|\mathrm{R}\rangle$

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SCHEMATICS OF OUR MODEL

3-level system : $\mathcal{H} = \mathrm{Vect}(|0\rangle, |\mathrm{L}\rangle, |\mathrm{R}\rangle)$



Continuous measurement of the electron position

System Evolution

We consider the density matrix of the system conditioned on the (stochastic) measurement results:

$$\mathrm{d}\rho_t = \mathrm{d}\rho_t^{\text{tunnel}} + \mathrm{d}\rho_t^{\text{bath}} + \mathrm{d}\rho_t^{\text{measure}}$$

Unitary tunnelling

$$d\rho_t^{\text{tunnel}} = -i[H, \rho_t]dt$$

Unitary tunnelling

$$d\rho_t^{\text{tunnel}} = -i[H, \rho_t]dt$$

with a simple **hopping** Hamiltonian:

$$H = \frac{\omega}{2}(|L\rangle\langle R| + |R\rangle\langle L|)$$

Dissipative coupling to the baths

$$\mathrm{d}\rho_t^{\mathrm{bath}} = \mathcal{L}^{\mathrm{bath}}(\rho_t)\,\mathrm{d}t$$

with:

$$\mathcal{L}^{\mathrm{bath}}(\rho_t) = \left(\mathsf{L}_{\sigma_l^+} + \mathsf{L}_{\sigma_l^-} + \mathsf{L}_{\sigma_r^+} + \mathsf{L}_{\sigma_r^-}\right)(\rho_t)$$

Dissipative coupling to the baths

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Here

$$\mathsf{L}_\sigma(\rho) = \sigma\rho\sigma^\dagger - \frac{1}{2}\{\sigma^\dagger\sigma, \rho\}$$

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Here

$$\mathsf{L}_\sigma(\rho) = \sigma\rho\sigma^\dagger - \frac{1}{2}\{\sigma^\dagger\sigma, \rho\}$$

with $\sigma_l^+ = \sqrt{a}e^{\beta\mu_l/2}|L\rangle\langle 0|$, $\sigma_l^- = \sqrt{a}|0\rangle\langle L|$, $\sigma_r^+ = \sqrt{b}e^{\beta\mu_r/2}|R\rangle\langle 0|$ and $\sigma_r^- = \sqrt{b}|0\rangle\langle R|$ such that we get Gibbs in the absence of tunnelling and measurement.

Continuous position measurement

$$d\rho_t = \mathcal{L}(\rho_t)dt + \underbrace{\gamma^2 L_{\mathcal{O}}(\rho_t)dt + \gamma\, D_{\mathcal{O}}(\rho_t)dW_t}_{\text{measurement part}}$$

with the stochastic innovation part of the measurement :

$$D_{\mathcal{O}}(\rho_t) = \mathcal{O}\rho_t + \rho_t\mathcal{O} - 2\,\mathrm{tr}(\mathcal{O}\rho_t)\rho_t$$

and the deterministic part :

$$L_{\mathcal{O}}(\rho_t) = \mathcal{O}\rho_t\mathcal{O}^\dagger - \frac{1}{2}\left\{\mathcal{O}^\dagger\mathcal{O}, \rho_t\right\}$$

Position measurement $\Rightarrow \mathcal{O} = \lambda_0|0\rangle\langle 0| + \lambda_L|L\rangle\langle L| + \lambda_R|R\rangle\langle R|$

Measurement Results

$$dX_t = 2\,\mathrm{tr}(\mathcal{O}\rho_t)dt + dW_t$$

In a DQD, X_t is the total charge going through the QPC.

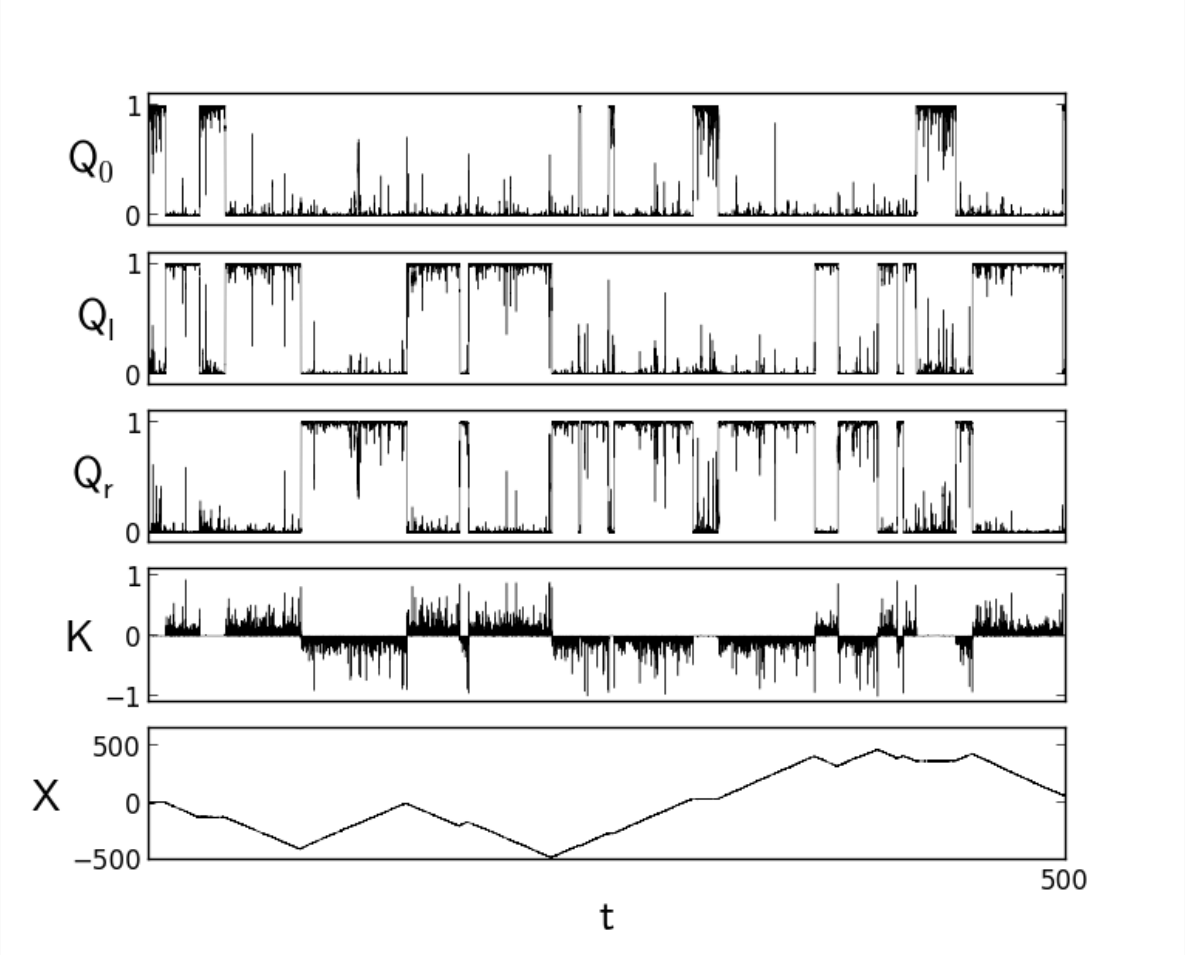
Experimentalists tend to consider the **intensity**, which is a smoothed version of the singular $\frac{dX_t}{dt}$.

Parametrisation

A simple and **stable** parametrisation of the density matrix is the following:

$$\rho = \begin{bmatrix} Q_0 & 0 & 0 \\ 0 & Q_l & iK/2 \\ 0 & -iK/2 & Q_r \end{bmatrix}$$

SNAPSHOT OF A TRAJECTORY

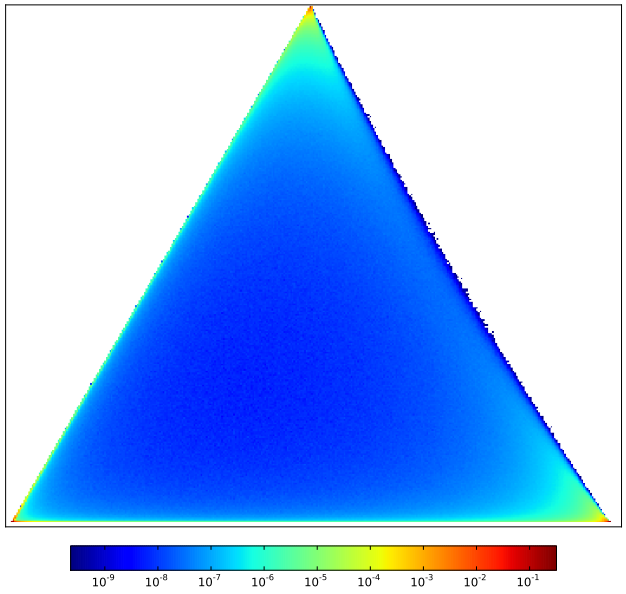


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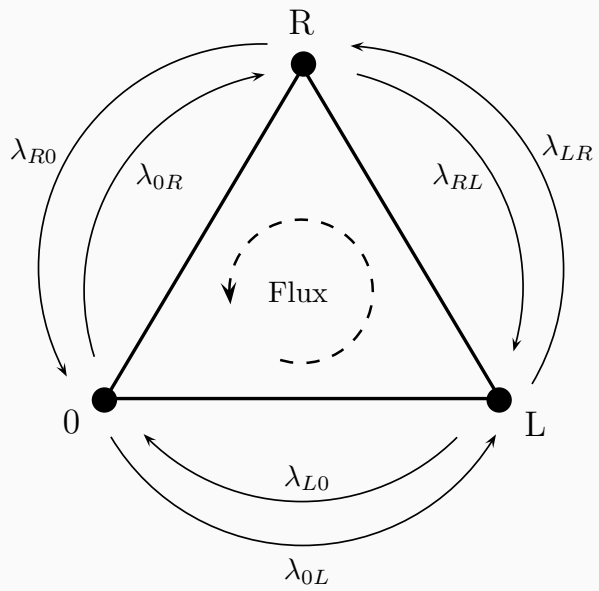
JUMPS AND FLUX

EQUIVALENT MARKOV PROCESS

Classically inspired definition of the electron flux:



$\longrightarrow$ Markov



$$\langle J \rangle_{stat} = \lim_{t \to +\infty} \frac{N_{L \to R}(t) - N_{R \to L}(t)}{t}$$

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Master equation:

$$\begin{cases} \partial_t \Pi_0 = -\left(\lambda_{0L} + \lambda_{0R}\right) \Pi_0 + \lambda_{R0} \Pi_R + \lambda_{L0} \Pi_L \\ \partial_t \Pi_L = -(\lambda_{L0} + \lambda_{LR}) \Pi_L + \lambda_{RL} \Pi_R + \lambda_{0L} \Pi_0 \\ \partial_t \Pi_R = -(\lambda_{RL} + \lambda_{R0}) \Pi_R + \lambda_{LR} \Pi_L + \lambda_{0R} \Pi_0 \end{cases}$$

Master equation:

$$\begin{cases} \partial_t \Pi_0 = -\left(\lambda_{0L} + \lambda_{0R}\right) \Pi_0 + \lambda_{R0} \Pi_R + \lambda_{L0} \Pi_L \\ \partial_t \Pi_L = -(\lambda_{L0} + \lambda_{LR}) \Pi_L + \lambda_{RL} \Pi_R + \lambda_{0L} \Pi_0 \\ \partial_t \Pi_R = -(\lambda_{RL} + \lambda_{R0}) \Pi_R + \lambda_{LR} \Pi_L + \lambda_{0R} \Pi_0 \end{cases}$$

Average statistical flux:

$$\langle J \rangle_{stat} = \lambda_{LR} \Pi_L^{stat} - \lambda_{RL} \Pi_R^{stat}$$

We just have to compute the λ's

Jump rates

$$\begin{cases} \lambda_{0L} = a e^{\beta \mu_l}, \ \lambda_{L0} = a \\ \lambda_{0R} = b e^{\beta \mu_r}, \ \lambda_{R0} = b \\ \lambda_{LR} = \lambda_{RL} \simeq \dfrac{\omega^2}{\gamma^2} \end{cases}$$

The different jump mechanisms decouple.

FLUX

Compare with the standard quantum mechanical flux

$$\langle \mathrm{J} \rangle_{\mathrm{QM}} = \mathrm{tr}\left(\hat{\mathrm{J}}\rho_{\mathrm{stat}}\right) \longrightarrow \text{identical for large } \gamma$$

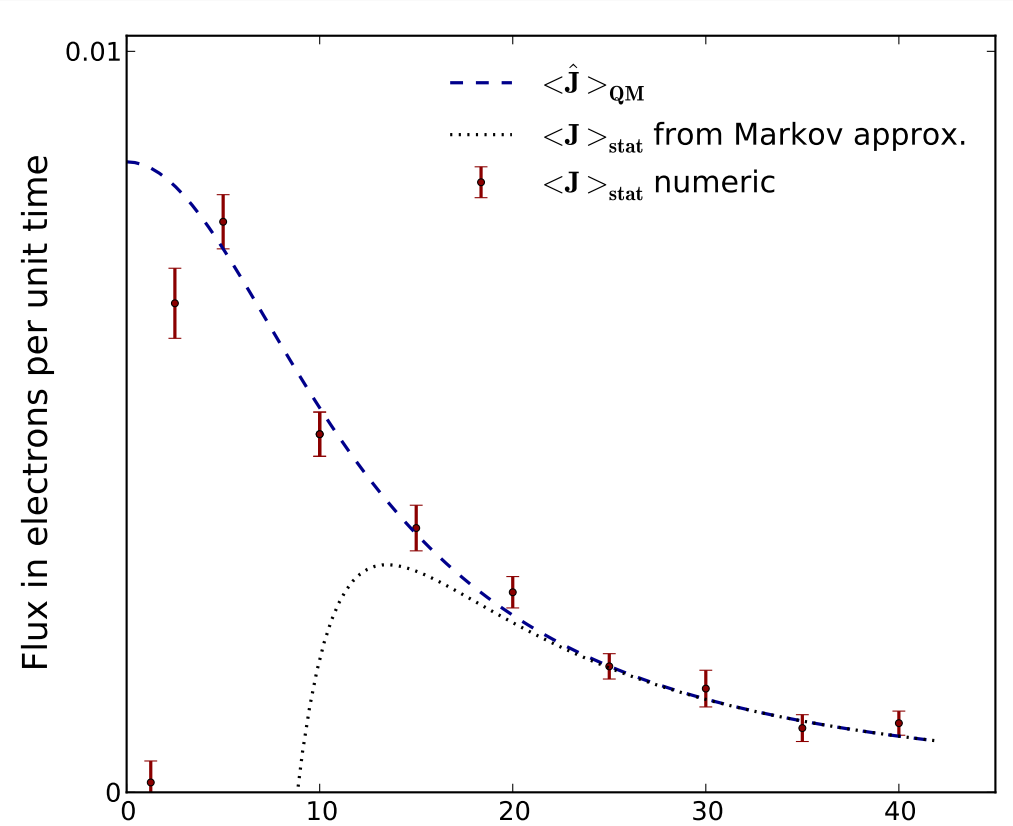


Figure: Flux as a function of γ

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FEEDBACK

Minimal feeedback ?

- Use the knowledge of X_t to compute ρ_t. Can be done exactly of approximately reading the **measurement result average slope**.
- Try to control the system using measurement backaction, i.e. no additionnal Hamiltonian.
- Only play with the measurement **rate** $\gamma(t)$, i.e. not the pointer states.

$\Rightarrow$ If you can measure a quantum system, you can probably use this feedback scheme.

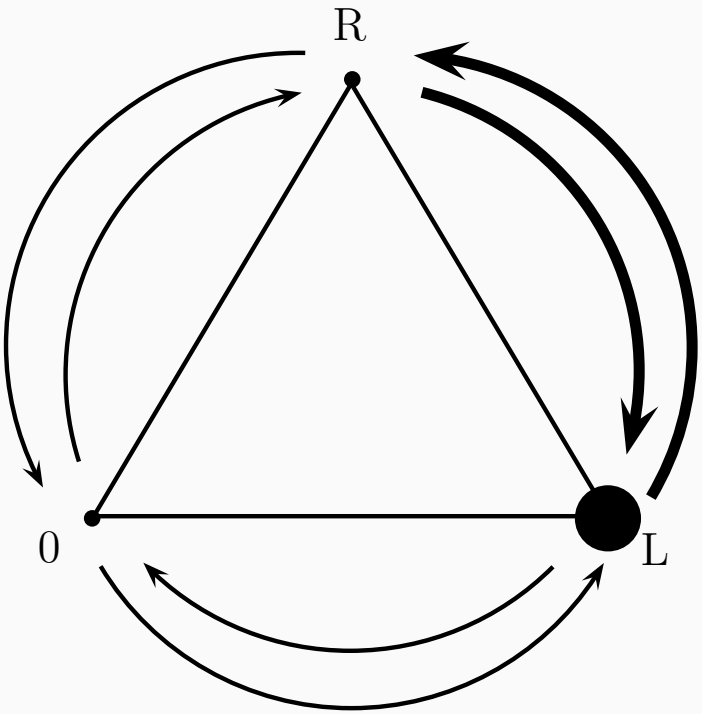
IDEA

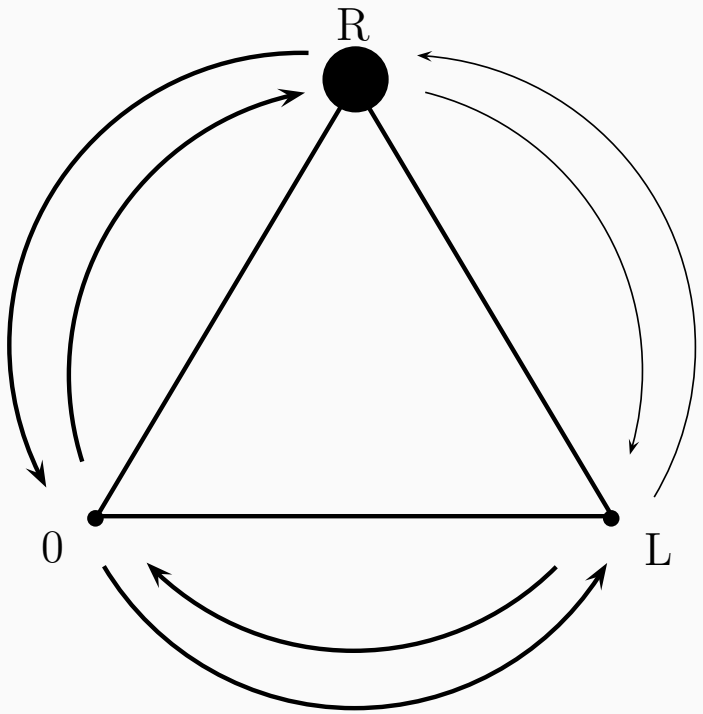
Control the $\mathbf{e}^-$-flux going through the DQD

Exploit the fact that the jump rates do not have the same dependency in γ.

- Measure strongly when the electron is likely to be in the right dot.
- Measure mildly when the electron is likely to be in the left dot.

IDEA



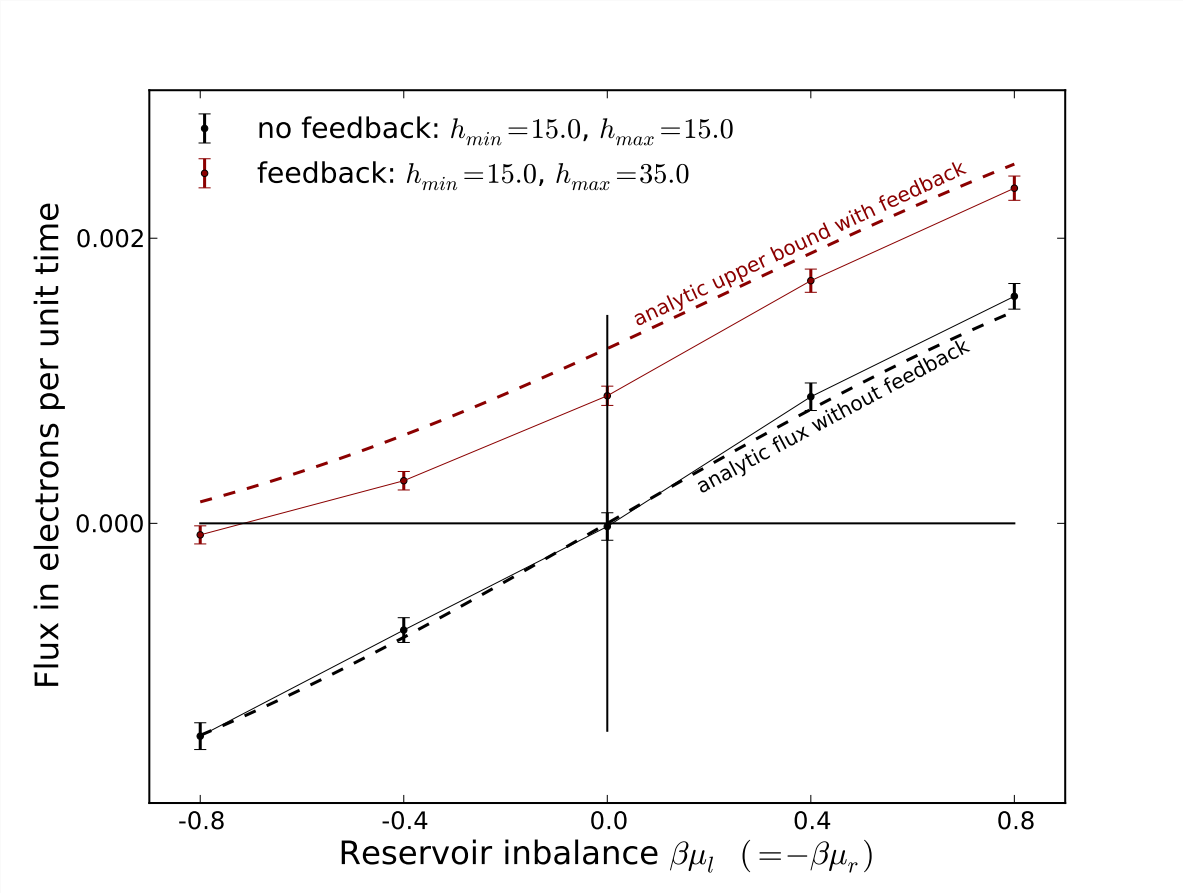


e^- on the left, very large γ

e^- on the right, not so large γ

RESULTS

$$\langle J \rangle_{\text{stat}} \propto \omega^2 \left(\frac{e^{\beta \mu_l}}{\gamma^2_{\text{min}}} - \frac{e^{\beta \mu_r}}{\gamma^2_{\text{max}}} \right)$$



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Consequences

- Possible to counter an arbitrary difference of potential
- Quantum Maxwell Daemon using **measurement only**.

CONCLUSION

SUMMARY

- Strong measurements give rise to a jumpy behaviour
- It is possible to redefine quantum mechanical quantities in a classical way using these jumps.
- The explicit dependence on measurement intensity allows for a simple control.

Not that crazy: We do not control the system by blinking but by playing with the light switch. Oscillating between to symmetric situations can produce work $\simeq$ molecular motors, ratchets, etc.

Other facts

- The scheme still works with other (non monitored) decoherence channels
- The scheme still works (numerically) for γ not so large

About applications

- Naive application to the control of flux in DQD's (probably useless)
- Detection of possible biases in adaptive measurement schemes
- Applications... to theory building

WHAT NEXT ?

- Find a group that can implement this scheme (in DQDs or in perhaps in cQED).
- Measure quantities mixing the system and the bath using quantum noises
- Say more generally what can be controlled for a given measure (anything that does not commute with the measured observable ?)
- Develop a **quantum stochastic thermodynamics** theory where this example could be consistently described without paradox.